



*Let the mind manage the body
Que l'esprit gère le corps*

**MAURITIUS
EXAMINATIONS
SYNDICATE**

**NCE 2020-2021
GRADE 9
Technology (Food &
Textiles)**

Subject code: N550

EXAMINERS' REPORT

April 2022

KEY MESSAGES

- It was encouraging to note that the majority of candidates attempted all questions and demonstrated a sound knowledge of the learning outcomes expected at this level.
- Candidates should be prepared by ensuring that they have full understanding of the requirements of different question types: for example, *name*, *state*, *discuss* and *explain*. Often, candidates had difficulty in obtaining the full marks for questions asking for an explanation because answers were not supported with relevant examples and reasoning.
- Generally candidates responded well to Section A of the question paper but found Section B more challenging.

GENERAL COMMENTS

The National Certificate of Education (NCE) Assessment comes at the end of the Nine-Year-Continuous-Basic-Education and is the second important national assessment that candidates take after the Primary School Achievement Certificate (PSAC). As with the PSAC assessment, the NCE assessment is based on the National Curriculum Framework Grades 7, 8 and 9 (*MIE, 2017*) and the Teaching and Learning Syllabus Grades 7, 8 and 9 (*MIE, 2020*).

The purpose of the Technology Studies Assessment in the context of the nine-year Continuous Basic Education is to assess the understanding of a range of concepts and skills related to Design & Technology and Food & Textiles Studies.

In Food and Textiles Studies, the focus of the learning is on three main areas:

- Nutrition and Health
- Self And Family Awareness
- Textile Technology

Food and Textiles Studies provides opportunities to **ALL** adolescents at the lower secondary, irrespective of gender, to develop a unique repertoire of knowledge, practices and dispositions that will assist them in meeting the challenges of everyday life in our evolving society.

The assessment objectives and their respective weighting is given in Table 1 (Annual Programme for the National Certificate of Education Assessment 2020-2021, *MES 2021*):

Table 1: Assessment Objectives

Assessment Objectives	Weighting %	
	Design and Technology (D&T)	Food and Textiles Studies (FTS)
Knowledge and Understanding	25	25
Application Skills	20	20
Analysis and Evaluation	5	5

The NCE assessment in Food and Textiles Studies focuses on the learning outcomes given at Grade 9 in the Teaching and Learning syllabus.

The NCE assessment 2020-2021 took place in special circumstances with the COVID-19 pandemic and its ensuing disruptions to school life and the school calendar. Due to these disruptions, the assessment was based on only part of the syllabus at Grade 9. Table 2 gives the learning outcomes which were not assessed in the 2020-2021 assessment.

A significant part of the chapter ‘Design and Creativity through textiles’ was not assessed. It was expected that this measure would help students focus on certain parts of the syllabus which were considered essential for them to progress to continue their learning in Fashion and Textiles Studies at Grade 10.

Table 2: Unit not assessed

UNIT	SPECIFIC LEARNING OUTCOMES (LOs) OF TLS NOT TO BE ASSESSED
Unit 8: Design and Creativity through Textiles - Textile Decoration Techniques	• List the appropriate resources required to create textile items. • Identify ways to recycle textile materials. • Design and make textile items using selected fabric decoration and coloration techniques. • Demonstrate safe practices when constructing a textile item. • Create textile items by applying basic elements of fashion design.

Given the above, the weightings of the different assessment objectives were slightly adjusted to cater for the significant disruptions in learning.

The question paper consisted of 10 questions assessing the 12 main topics, that is;

- Micronutrients and Macronutrients
- Food habits and food choices
- Methods of cooking
- Convenience foods
- Building strong family relationship
- Family and Technology

- Sustainable food production and consumption practices
- Consumer Awareness – Care Labels
- Fabric constructions, their performance characteristics and end-uses
- Smart and Modern Fabrics
- Elements of Fashion design
- Different parts of an electric sewing machine/ Safety precautions while using an electric sewing machine

The Food and Textiles Studies paper produced a good spread of marks with some candidates scoring in the top mark range. However, some candidates had difficulties with Question 1 (c), (d) and (e), Question 9 and Question 10.

SPECIFIC COMMENTS

SECTION A

Question 1- (Multiple Choice Question)

Question 1 consisted of 6 multiple-choice questions assessing mainly AO1, Knowledge and Understanding.

The items which proved to be within the reach of most candidates were (a) and (b). For the other items, less than 7 out of 10 candidates found the correct answer. It seemed that candidates did not read the items attentively enough, especially for items (d), (e) and (f).

Table 3 provides the answer key for Question 1.

Table 3: Answer keys for Question 1

Part	Key
1	C
2	A
3	B
4	A
5	D
6	B

Comments on specific parts

Item 1: *The narrow finished edge of a woven fabric is known as a*

This item was knowledge-based and candidates had to recall the information. They had to understand the term ‘woven’ first to choose the correct example. The correct answer C ‘Selvedge’ was provided by most candidates.

Item 2: *One of the basic elements of fashion design is*

The correct answer is A ‘Colour’.

This part proved to be challenging for a significant number of candidates although it tested simple recalling of information. The three distractors were equally popular among the wrong answers.

Item 3: *The should be lowered before starting to stitch.*

About two-thirds of candidates found the correct answer B ‘Presser foot’. However, a large number of candidates opted for ‘thread take-up’.

The learning outcome ‘Identify the parts of an electric sewing machine and how to use the electric sewing machine properly and safely’ was being assessed here. Besides, it is important to know how to operate an electric sewing machine and their safety aspects before using it.

Item 4: *The stitching shown in the Figure is known as a*

This item was based on a picture showing embroidery stitch which candidates learnt in Grade 9. Candidates were requested to name the Embroidery Stitch.

A majority of candidates found the correct answer: A, ‘loop’. A non-negligible number gave option D ‘Hemming’.

Performance on this part was satisfactory. Very often candidates confused between sewing stitches and embroidery stitches.

Item 5: *The silhouette illustrated in Figure 2 is known as*

The correct answer is D ‘A-line’.

In this part, candidates must first understand these different terms ‘triangular’, ‘wedge’, ‘tubular’ and ‘A-line’ before answering the question. Candidates should demonstrate an understanding of the nature of fashion design.

Item 6: *Knitted fabrics are*

This item was relatively well answered by the majority of candidates. Candidates recognized the different fabrics performance characteristics related to fabric construction and their end-uses. The correct answer is B ‘warm’.

Question 2- (True/False)

Comments on different parts of Question 2

Part (a): *Fat soluble vitamins are stored under the skin.*

The correct answer is 'True'.

Almost all the candidates have been able to tackle this question correctly. This question is based on the topic Micronutrients – Vitamins, where candidates classified the water-soluble and fat-soluble vitamins and their functions.

Part (b): *Vitamin D prevents the growth of strong bones and teeth.*

The correct answer is 'False'.

This question has been answered correctly by the majority of the candidates. For those who gave wrong answers, there might have been some confusion while reading the item.

Students are introduced to the concept of the importance of nutrients – vitamins in Grade 6, 7 and 8 so as to be aware of different nutrients which are essential for the body.

Part (c): *Citrus fruits are good sources of vitamin C.*

The correct answer is 'True'.

Candidates were able to recognize the different types of fruits and their importance in the diet. They understood the meaning of citrus fruits. However, from wrong answers given by candidates, it seemed that they might have confused with other types of fruits such as 'kiwi', 'apple' and others.

Part (d): *Goitre is due to lack of iron in the body.*

The correct answer is 'False'.

The item was knowledge-based and candidates had to recall the information. They had to understand the term 'Goitre' first to be able to give the correct answer. The correct answer 'False' was provided by most candidates.

Question 3 (Matching)

This was a matching question with different care symbols given in Column A to be matched with their meanings and functions in Column B.

This was a highly scoring question attempted by most candidates. It showed a good knowledge of care symbols. Most candidates scored full marks for this question.

Item (1):



Among those who gave the wrong answer, 'Dry in shade' was the most common.

Item (2):



Many candidates identified this care symbol and its meaning correctly.

Item (3):



This care symbol was correctly identified by the majority of the candidates.

Item (4):



Most candidates correctly labelled the care symbol with its proper meaning in column B.

Item (5):



Most candidates responded to this care symbol well as they understood the meaning of the cross on the care label. The given example might have also helped.

Overall performance for this question was satisfactory as the majority of the candidates were able to recognize the importance of care symbols in textile items.

Question 4

This question was generally well answered. Many candidates were able to correctly name the method of cooking.

Item (a): *Cooking food in water at 100°C*

The correct answer is 'boiling'

The item is knowledge-based and candidates had to recall the information. Candidates were able to differentiate between the different methods of cooking among which one of them is boiling. The method boiling was identifiable by the reference to 100°C.

The majority of the candidates' answers to this question were correct.

Item (b): *Cooking food in hot oven*

The correct answer is 'baking'

Candidates had good knowledge and understanding of using the oven and they were aware of the main methods of cooking in an oven which included baking, roasting and grilling.

Most candidates responded well to this question.

Item (c): *Cooking food in hot oil*

The correct answer is 'frying'. It was well attempted by the majority of the candidates.

Item (d): *Cooking food using intense heat*

The correct answer is ‘grilling’.

Many candidates managed to associate the term ‘*grilling*’ to the use of intense heat for cooking food. Some wrong answers included baking, it might be because the oven is also used for grilling at home / school.

Item (e): *Cooking food in water vapour*

The correct answer is ‘steaming’.

Most candidates were able to name the method of cooking food in water vapour which is steaming. However, a few candidates gave wrong answers such as ‘boiling’ instead of steaming. Perhaps they did not differentiate between water and water vapour.

Question 5 - (Fill in the blanks)

For this question, candidates were able to fill in the blanks with correct answers as they were able to understand the importance of building strong family relationships, to distinguish between the positive and negative impacts of technology on family life and sustainable food production and consumption practices.

Part (a): *Addiction to technology involves the use of mobile devices.*

The correct answer is ‘obsessive’

This question was well answered by most candidates. They were able to grasp the impact of technology on family life, whether it is positive or negative.

Part (b): *One of the negative impacts of technology on family life is..... .*

The correct answer is ‘laziness’

Most of the candidates responded to this question well. They understood that negative impacts of technology on family life may lead to laziness, separation, dangers/risks, addiction/dependency and limited outdoor activities.

Part (c): *When going shopping, always carry bags with you.*

The correct answer is ‘eco-friendly’

The majority of students answered to this question well. Candidates had a good knowledge and understanding of the term ‘Go Green’, that is, the measures to be taken to prevent the different types of pollution in the world.

Part (d): *Sustainable food production and practices help to save energy, water and protect the environment.*

The correct answer is ‘consumption’

Candidates were able to differentiate between sustainable and unsustainable food production and consumption practices. This question was very well answered by almost all the candidates.

SECTION B

Question 6 - (Non-communicable Diseases)

Part (a): *Name THREE Non-Communicable Diseases (NCDs) which may result due to unhealthy food habits.*

Most candidates gave at least one correct answer of the NCDs. High achievers provided the correct answers for all three NCDs. The most common correct answers were cancer, heart diseases, diabetes and cardiovascular diseases.

Some candidates confused between NCDs and eating disorders, diseases or symptoms.

Part (b): *State TWO factors that adolescents need to consider to ensure that the food is safe when eating out.*

This part was mostly well answered by many candidates. Despite the fact that some candidates faced difficulties to answer in full sentences, they were able to show understanding of this part of the topic.

However, there were some candidates who provided incomplete or vaguely written statements. Some did not attempt this part at all. Often keywords were missing in answers. This prevented candidates from obtaining full marks.

Question 7 - (Convenience foods)

Part (a): *Give TWO examples of convenience food.*

Candidates generally offered appropriate examples of convenience foods. A wide array of acceptable answers was obtained. Most of the candidates were able to answer this part well.

Some candidates gave types of convenience foods instead of examples.

Incomplete answers were also noticed, for e.g., ‘Beans’ instead of canned beans /baked beans. Brand names, such as KFC and Mc Donald, were also given and were not accepted.

Part (b): *State ONE advantage and ONE disadvantage of using convenience foods.*

Many candidates were able to give relevant advantages and disadvantages of convenience foods. They understood the term ‘convenience foods’.

A few candidates did not attempt this part or provided incomplete answers.

It was also noted that some candidates confused between convenience foods and fast foods as shown in part (a) above.

Question 8 - (Water)

Part (a): *State TWO importance of water in the diet of living things.*

On average, this part was well answered. Most candidates were able to correctly state the importance of water in the diet of living things.

Many candidates stated that water is important to ‘*stay alive*’ as per the statement provided.

Part (b): *List the symptoms of dehydration.*

Many candidates were able to correctly list the symptoms of dehydration. Some candidates made references to common Mauritian beliefs instead of demonstrating sound content knowledge as per the syllabus content.

Part (c): *Give TWO ways to avoid dehydration.*

The majority of candidates scored full marks for giving different ways to avoid dehydration as they were familiar with the term ‘dehydration’ and its meaning.

Candidates should be advised to include the number of glasses/litres of liquid to be consumed in order to score full marks.

Question 9 - (Modern/Smart fabrics)

Few candidates could correctly attempt both parts (a) and (b).

Part (a): *Kevlar is a modern fabric used to make parachutes. List TWO properties of Kevlar that make it suitable for making parachutes.*

Some candidates were precise in their answers and demonstrated clear understanding of the properties of ‘Kevlar’. However, many candidates did not attempt this question successfully.

Some candidates answered that ‘Kevlar is resistant’ but did not specify to what, thus lacking precision in their answers. Some candidates confused the term ‘properties’ with ‘uses’.

Part (b): *Give TWO uses of reflective textiles.*

The majority of candidates could not state the two uses of reflective textiles, but provided at least one appropriate use. However, the answers given for this part were quite varied but few answers were specific and related to the context.

Some candidates did not understand the term ‘reflective textiles’ properly.

Question 10- (An electric sewing machine)

Part (a): *Figure 3 shows an electric sewing machine. Label parts A, B, C, and D*

Few candidates scored full marks for part (a).

Many candidates managed to score some marks. Some common mistakes included 'presser foot' instead of 'foot pedal' and 'bobbin winder' instead of 'bobbin'. However, many candidates used the words given in Question 1(c) to label the parts of the sewing machine.

Part (b): *Give THREE safety precautions that you should observe when using an electric sewing machine.*

Many candidates demonstrated poor content knowledge for this question. Candidates failed to specify the safety precautions when using the sewing machine but they were familiar on how to use the sewing machine.

It was also observed that many answers were related to the prior preparation to sewing *instead* of the safety precautions when using the sewing machine. Some of the answers provided for this part were not clear. It was difficult to clearly understand what the candidates meant, especially when there were wrong uses of technical words and words wrongly written and ideas unclear.

Some answers provided were not appropriate for operating a sewing machine but the answers were rather applicable for using tools/machines in the workshop, for e.g., use of goggles, gloves and protective clothing/equipment.

Recommendations:

Some recommendations to improve performance:

- Candidates should be encouraged to write legibly and clearly.
- Candidates should read questions carefully before attempting to answer them.
- Candidates should read and understand the instructions. They can underline key words/technical terms where applicable.
- Teaching and learning should be in line with the Teaching and Learning Syllabus of 'Technology Studies: Food and Textiles Studies'.
- Candidates must be encouraged to explain their reasoning, especially when the question requires longer pieces of writing.
- More practice on how to interpret a given situation and answer in a concise manner is desirable.
- Candidates must be more familiar in using Food and Textiles Studies terms in labelling of diagrams.
- Language skills must be improved. These skills are essential for students to learn and understand the subject as clearly described in their Food and Textiles Studies syllabus and these are also transferrable skills which apply to other subject areas.